SET 22: TIME SERIES ANALYSIS

Dataset	Monthly Sales Data
Department	AIML
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Day	Thursday

Dataset: Monthly Sales Data

Month	Sales (in $)
January	15000
February	18000
March	22000
April	20000
May	23000

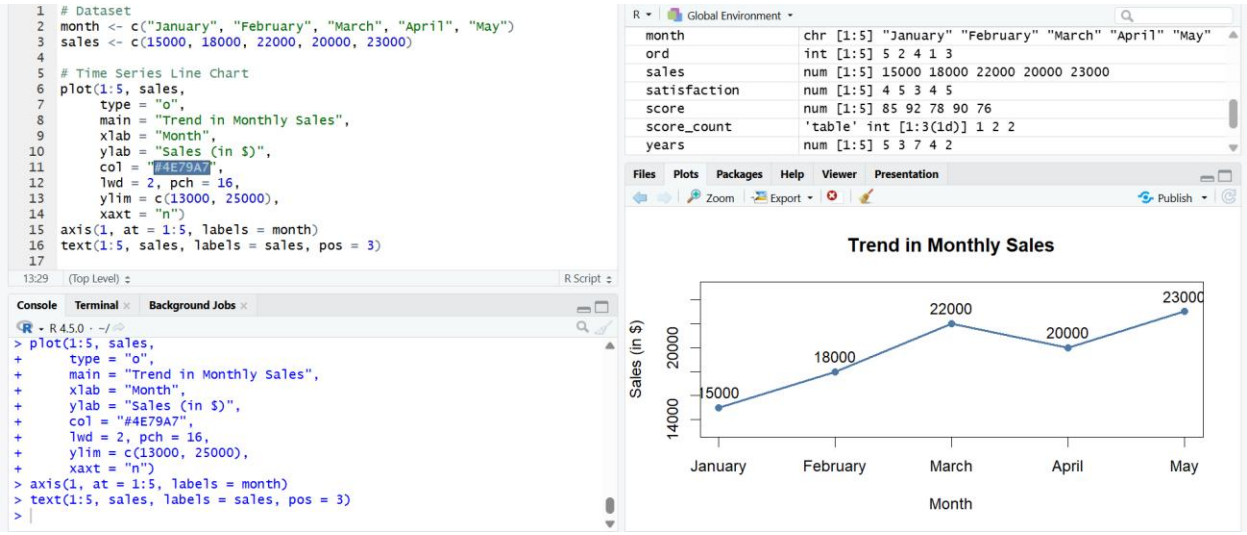
PROGRAM 1: Time Series Line Chart of Monthly Sales

Question: In R, create a time series line chart to visualize the trend in monthly sales. Label the axes and title the chart.

```
# Dataset
month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)

# Time Series Line Chart
plot(1:5, sales,
     type = "o",
     main = "Trend in Monthly Sales",
     xlab = "Month",
     ylab = "Sales (in $)",
     col = "#4E79A7",
     lwd = 2, pch = 16,
     ylim = c(13000, 25000),
     xaxt = "n")
axis(1, at = 1:5, labels = month)
text(1:5, sales, labels = sales, pos = 3)
```

Output 1: Monthly Sales Line Chart



Observation: Sales rose from $15,000 in January to $22,000 in March, dipped to $20,000 in April and reached the highest value of $23,000 in May. Overall there is an upward trend of $8,000 (about 53%) over the five months.

PROGRAM 2: Scatter Plot of Advertising Budget vs Monthly Sales

Question: In R, generate a scatter plot to analyze the relationship between advertising budget and monthly sales. Explain any insights.

Note: this data is not given in the question, so sample values are assumed for the demonstration. Replace them with your actual data if you have it.

Month	Advertising Budget ($)	Sales ($)
January	2000	15000
February	2300	18000
March	3200	22000
April	3000	20000
May	3300	23000

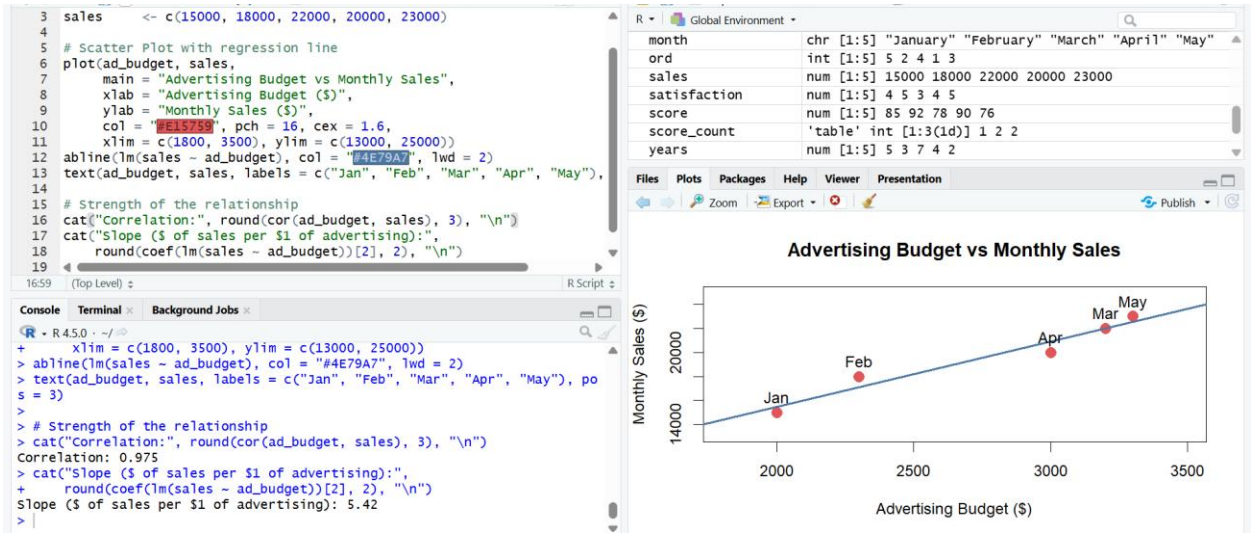
```
# Dataset (advertising budget values are assumed sample data)
ad_budget <- c(2000, 2300, 3200, 3000, 3300)
sales     <- c(15000, 18000, 22000, 20000, 23000)

# Scatter Plot with regression line
plot(ad_budget, sales,
     main = "Advertising Budget vs Monthly Sales",
     xlab = "Advertising Budget ($)",
     ylab = "Monthly Sales ($)",
     col = "#E15759", pch = 16, cex = 1.6,
     xlim = c(1800, 3500), ylim = c(13000, 25000))
abline(lm(sales ~ ad_budget), col = "#4E79A7", lwd = 2)
```

```
text(ad_budget, sales, labels = c("Jan", "Feb", "Mar", "Apr", "May"), pos =
3)

# Strength of the relationship
cat("Correlation:", round(cor(ad_budget, sales), 3), "\n")
cat("Slope ($ of sales per $1 of advertising):",
    round(coef(lm(sales ~ ad_budget))[2], 2), "\n")
```

Output 2: Advertising Budget vs Monthly Sales Scatter Plot




```
> R Console output
Correlation: 0.975
Slope ($ of sales per $1 of advertising): 5.42
```

Observation: The points rise from left to right and lie close to the fitted line, giving a correlation of 0.975, i.e. a strong positive relationship. The slope shows that each extra $1 of advertising is associated with about $5.42 of additional sales. The months with the largest budgets (March and May) had the highest sales. Insight: higher advertising spend goes together with higher sales, but with only five points, and with assumed budgets, this shows association, not proof of cause.

PROGRAM 3: Tableau Dashboard: Sales Trend and Product Distribution

Question: In Tableau, build a dashboard combining a line chart showing sales trend and a pie chart displaying the distribution of sales across products.

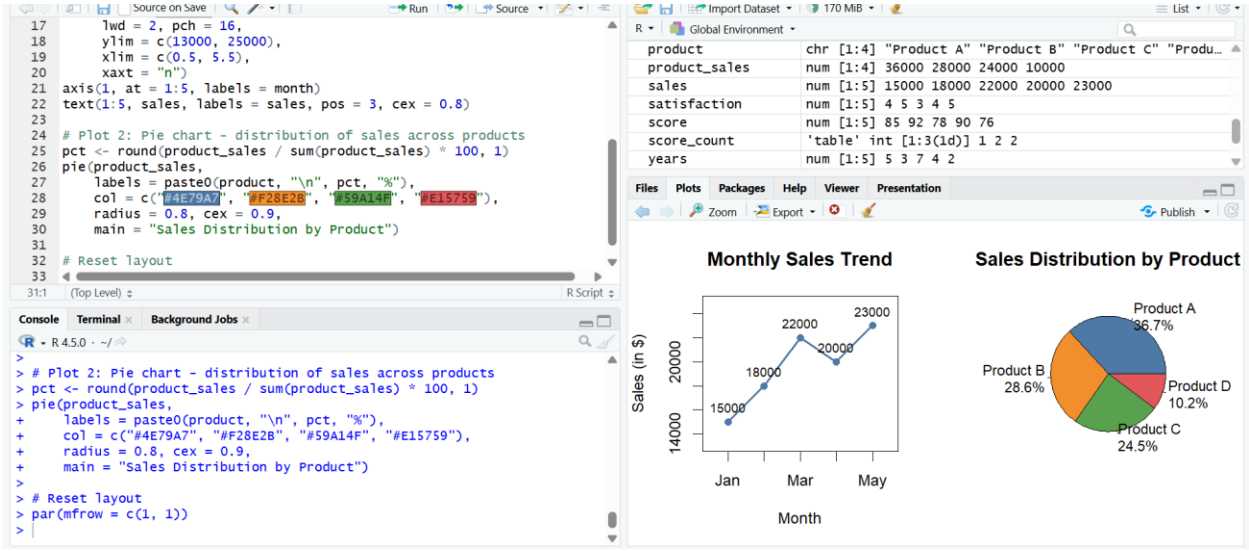
Note: the product-wise sales are not given in the question, so sample values are assumed (they add up to the same total of $98,000 as the monthly sales).

Product	Sales ($)	Share of Total
Product A	36000	36.7%
Product B	28000	28.6%
Product C	24000	24.5%
Product D	10000	10.2%

Tableau Steps

1. Open Tableau and connect to the data file (Excel/CSV) containing the Monthly Sales data and the Product Sales data.
2. Verify the fields: Month should be a Date (or Discrete text) field, Sales a numeric measure and Product a dimension.
3. Create a worksheet named “Sales Trend”. Drag Month to Columns and Sales to Rows (SUM), and select Line in the Marks card.
4. Add the title “Monthly Sales Trend”, label the axes (Month, Sales ($)) and show mark labels.
5. Create a worksheet named “Sales by Product”. In the Marks card choose Pie, drag Product to Color and Sales to Angle.
6. Drag Product and Sales to Label; right-click SUM(Sales) > Quick Table Calculation > Percent of Total so each slice shows its percentage. Add the title “Sales Distribution by Product”.
7. Create a new Dashboard, set the size to Automatic, and drag in the “Sales Trend” and “Sales by Product” sheets.
8. Add the title “Monthly Sales Dashboard”, and click the “Use as Filter” icon on the pie chart; add a Month quick filter (Show Filter).
9. Test the dashboard by clicking a pie slice or choosing months in the filter and checking that the views update.

Output 3: Monthly Sales Dashboard (Tableau)



Observation: The dashboard shows the upward sales trend from January to May next to the product mix. Product A contributes the largest share of sales (36.7%) and Product D the smallest (10.2%). The filters let the user explore any month or product interactively.

PROGRAM 4: Autocorrelation Plot of Monthly Sales

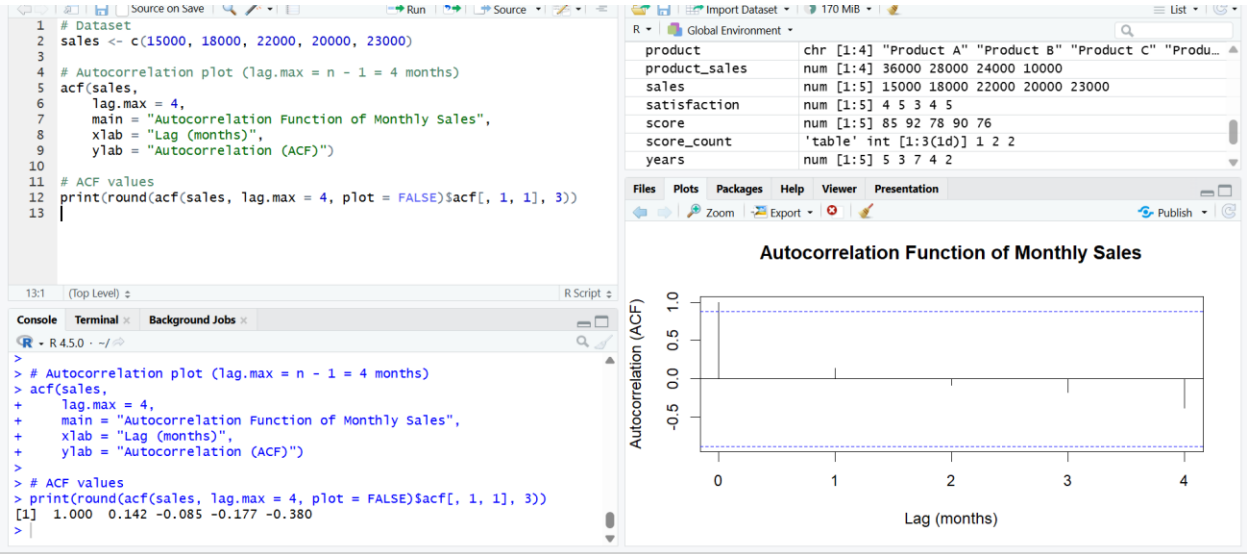
Question: In R, create an autocorrelation plot to identify seasonality in the time series data.

```
# Dataset
sales <- c(15000, 18000, 22000, 20000, 23000)

# Autocorrelation plot (lag.max = n - 1 = 4 months)
acf(sales,
    lag.max = 4,
    main = "Autocorrelation Function of Monthly Sales",
    xlab = "Lag (months)",
    ylab = "Autocorrelation (ACF)")

# ACF values
print(round(acf(sales, lag.max = 4, plot = FALSE)$acf[, 1, 1], 3))
```

Output 4: Autocorrelation (ACF) Plot




```
> R Console output
[1]  1.000  0.142 -0.085 -0.177 -0.380
```

Observation: The autocorrelations at lags 1 to 4 are 0.142, -0.085, -0.177 and -0.380, and all of them lie inside the dashed 95% confidence bands (about ±0.88), so no significant autocorrelation or repeating seasonal spike is visible. However, only five months of data are available, so seasonality cannot be reliably judged; at least two full seasonal cycles (for example 24 monthly values) are needed.

Procedure

1. Enter the monthly sales data in R and draw the time series line chart using plot() with type = "o".
2. Create the advertising budget data, draw the scatter plot with plot() and add a regression line with abline(lm()).

3. Calculate the correlation with cor() and interpret the relationship.
4. Draw the autocorrelation plot of the sales series using acf().
5. Open Tableau and import the Monthly Sales and Product Sales data.
6. Create the line chart and the pie chart as separate worksheets.
7. Combine the worksheets into a dashboard, add a title and enable filters.
8. Test the dashboard interactivity.

Result

The monthly sales data was successfully visualized using a time series line chart, a scatter plot of advertising budget against sales, an autocorrelation plot, and an interactive Tableau dashboard combining the sales trend and the product-wise sales distribution.